ClipSAT : Cambridge A2 Level Math 9709 [ANSWER KEY]

Unit 0.01: A2-Level 9709 Paper 3 & Paper 6 Diagnostic Readiness

Q1. Which of the following is the correct expression of dfrac{5x + 1}{x - 1x + 2} in partial fractions?

[x] A: dfrac{2}{x - 1} + dfrac{3}{x + 2}

[ ] B: dfrac{3}{x - 1} + dfrac{2}{x + 2}

[ ] C: dfrac{2}{x - 1} - dfrac{3}{x + 2}

[ ] D: dfrac{1}{x - 1} + dfrac{4}{x + 2}

Solution: To decompose the rational function into partial fractions, set up the identity with distinct linear factors: frac{5x + 1}{x - 1x + 2} equiv frac{A}{x - 1} + frac{B}{x + 2} Multiplying both sides by the common denominator x - 1x + 2: 5x + 1 equiv Ax + 2 + Bx - 1 To determine A, substitute x = 1: 51 + 1 = A1 + 2 + B0 implies 6 = 3A implies A = 2 To determine B, substitute x = -2: 5-2 + 1 = A0 + B-2 - 1 implies -9 = -3B implies B = 3 Therefore, the partial fraction decomposition is: frac{2}{x - 1} + frac{3}{x + 2} Common Misconceptions: - Choice B incorrectly associates A with the root x = -2 and B with x = 1, reversing the numerators. - Choice C results from a sign error when solving -9 = -3B. - Choice D assumes coefficients must sum directly to 5 without properly weighting by denominator factors. [M1][A1]

Q2. A curve has equation y = ln2x^3 + 1 . What is the gradient of the tangent to the curve at the point where x = 1 ?

[x] A: 2

[ ] B: 6

[ ] C: dfrac{2}{3}

[ ] D: ln 3

Solution: To find the gradient of the tangent to the curve y = ln2x^3 + 1 at x = 1 , we differentiate y with respect to x using the chain rule. Let u = 2x^3 + 1 , so y = lnu . Then: dfrac{mathrm{d}u}{mathrm{d}x} = 6x^2 quad text{and} quad dfrac{mathrm{d}y}{mathrm{d}u} = dfrac{1}{u} Applying the chain rule: dfrac{mathrm{d}y}{mathrm{d}x} = dfrac{mathrm{d}y}{mathrm{d}u} cdot dfrac{mathrm{d}u}{mathrm{d}x} = dfrac{6x^2}{2x^3 + 1} Now evaluate the derivative at x = 1 : left.dfrac{mathrm{d}y}{mathrm{d}x}right|_{x=1} = dfrac{61^2}{21^3 + 1} = dfrac{6}{3} = 2 Misconception Analysis: - Choice B 6 occurs if one differentiates only the inner function 2x^3 + 1 to get 6x^2 and evaluates it at x = 1 , neglecting the outer logarithmic derivative. - Choice C frac{2}{3} arises from an incorrect derivative of the inner polynomial, such as differentiating 2x^3 incorrectly as 2x . - Choice D ln 3 confuses the gradient of the tangent with the y -coordinate of the point on the curve, y1 = ln21^3 + 1 = ln 3 . [M1][A1]

Q3. Find the exact value of the definite integral int_{0}^{ln 2} mathrm{e}^{2x} , mathrm{d}x.

[x] A: dfrac{3}{2}

[ ] B: 3

[ ] C: dfrac{1}{2}

[ ] D: dfrac{7}{2}

Solution: To evaluate the definite integral: 1. Find the antiderivative using the standard standard result int mathrm{e}^{kx} , mathrm{d}x = frac{1}{k}mathrm{e}^{kx} + C: int mathrm{e}^{2x} , mathrm{d}x = frac{1}{2}mathrm{e}^{2x} 2. Apply the limits from x = 0 to x = ln 2: left[frac{1}{2}mathrm{e}^{2x} right]_{0}^{ln 2} = frac{1}{2}mathrm{e}^{2ln 2} - frac{1}{2}mathrm{e}^{20} 3. Simplify the exponential terms using the power law of logarithms, 2ln 2 = ln2^2 = ln 4: mathrm{e}^{2ln 2} = mathrm{e}^{ln 4} = 4 and mathrm{e}^{0} = 1 4. Compute the final value: frac{1}{2}4 - frac{1}{2}1 = 2 - frac{1}{2} = frac{3}{2} Common Errors: - Omitting the factor of frac{1}{2} gives 4 - 1 = 3 Choice B. - Forgetting that 2ln 2 = ln 4 and incorrectly setting mathrm{e}^{2ln 2} = 2 gives frac{1}{2}2 - 1 = frac{1}{2} Choice C. [M1][A1]

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Q4. The rational function f is defined for all real x notin {-1, 2} by fx = frac{x^3 + kx + 7}{x+1x-2^2} where k is a constant. The partial fraction decomposition of fx takes the form A + frac{B}{x+1} + frac{C}{x-2} + frac{D}{x-2^2} where A, B, C, and D are real constants. Given that C = 2 , find the value of k .

[x] A: -3
[ ] B: 1
[ ] C: 3
[ ] D: 24

Solution: Step 1: Determine the polynomial quotient A . The degree of the numerator is 3 and the degree of the denominator is 1 + 2 = 3 . Expanding the denominator gives: x+1x-2^2 = x+1x^2 - 4x + 4 = x^3 - 3x^2 + 4 Since the leading coefficient of both the numerator and the denominator is 1 , polynomial division yields a constant quotient: A = frac{1}{1} = 1 Step 2: Set up the algebraic identity. Multiplying both sides of the partial fraction expansion by x+1x-2^2 : x^3 + kx + 7 equiv Ax+1x-2^2 + Bx-2^2 + Cx+1x-2 + Dx+1 Step 3: Relate B and C by comparing coefficients of x^2 . On the left-hand side, the coefficient of x^2 is 0 . On the right-hand side, collecting the x^2 terms: - Ax+1x-2^2 = Ax^3 - 3x^2 + 4 implies -3A - Bx-2^2 = Bx^2 - 4x + 4 implies +B - Cx+1x-2 = Cx^2 - x - 2 implies +C - Dx+1 implies 0 Therefore: 0 = -3A + B + C Substitute A = 1 and the given value C = 2 : -31 + B + 2 = 0 implies B = 1 Step 4: Solve for k by substituting x = -1 . Setting x = -1 eliminates the terms containing x+1 : -1^3 + k-1 + 7 = B-1 - 2^2 -1 - k + 7 = B-3^2 6 - k = 9B Since B = 1 : 6 - k = 9 implies k = -3 Common Algebraic Traps: - k = 24 : Omits the quotient A treating an improper fraction as proper, giving B + C = 0 implies B = -2 , leading to 6 - k = -18 implies k = 24 . - k = 1 : Substitutes x = 1 instead of the root x = -1 , obtaining 1 + k + 7 = 9 implies k = 1 . - k = 3 : Makes a sign error when rearranging 6 - k = 9 . [M1][A1]

Q5. Which of the following is the correct expression of dfrac{5x + 1}{x - 1x + 2} in partial fractions?

[x] A: dfrac{2}{x - 1} + dfrac{3}{x + 2}
[ ] B: dfrac{3}{x - 1} + dfrac{2}{x + 2}
[ ] C: dfrac{2}{x - 1} - dfrac{3}{x + 2}
[ ] D: dfrac{1}{x - 1} + dfrac{4}{x + 2}

Solution: To decompose the rational function into partial fractions, set up the identity with distinct linear factors: frac{5x + 1}{x - 1x + 2} equiv frac{A}{x - 1} + frac{B}{x + 2} Multiplying both sides by the common denominator x - 1x + 2: 5x + 1 equiv Ax + 2 + Bx - 1 To determine A, substitute x = 1: 51 + 1 = A1 + 2 + B0 implies 6 = 3A implies A = 2 To determine B, substitute x = -2: 5-2 + 1 = A0 + B-2 - 1 implies -9 = -3B implies B = 3 Therefore, the partial fraction decomposition is: frac{2}{x - 1} + frac{3}{x + 2} Common Misconceptions: - Choice B incorrectly associates A with the root x = -2 and B with x = 1, reversing the numerators. - Choice C results from a sign error when solving -9 = -3B. - Choice D assumes coefficients must sum directly to 5 without properly weighting by denominator factors. [M1][A1]

Q6. The complex number z satisfies the equation z2 - mathrm{i} = 7 + 4mathrm{i}. Express z in the form x + mathrm{i}y , where x and y are real.

[x] A: 2 + 3mathrm{i}
[ ] B: dfrac{18}{5} + dfrac{1}{5}mathrm{i}
[ ] C: 2 - 3mathrm{i}
[ ] D: dfrac{10}{3} + 5mathrm{i}

Solution: To find z , rearrange the given equation: z = frac{7 + 4mathrm{i}}{2 - mathrm{i}} Multiply the numerator and the denominator by the complex conjugate of the denominator, which is 2 + mathrm{i} : z = frac{7 + 4mathrm{i}2 + mathrm{i}}{2 - mathrm{i}2 + mathrm{i}} Expand the denominator: 2 - mathrm{i}2 + mathrm{i} = 2^2 - mathrm{i}^2 = 4 - -1 = 5 Expand the numerator: 7 + 4mathrm{i}2 + mathrm{i} = 14 + 7mathrm{i} + 8mathrm{i} + 4mathrm{i}^2 = 14 + 15mathrm{i} - 4 = 10 + 15mathrm{i} Divide each term by the denominator: z = frac{10 + 15mathrm{i}}{5} = 2 + 3mathrm{i} Common Pitfalls: - Choice B occurs from mistakenly multiplying by 2 - mathrm{i} instead of the complex conjugate 2 + mathrm{i}. - Choice D arises from incorrectly evaluating the denominator as 2^2 - 1 = 3 rather than 4 - -1 = 5 . [M1][A1]

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Q7. A continuous random variable X has probability density function given by fx = frac{3}{8}x^2 for 0 le x le 2 , and fx = 0 otherwise. Find mathrm{P}1 le X le 2 .

[x] A: dfrac{7}{8}

[ ] B: dfrac{1}{8}

[ ] C: dfrac{3}{4}

[ ] D: dfrac{1}{2}

Solution: To find the probability mathrm{P}1 le X le 2 , we integrate the probability density function fx over the interval [1, 2] : mathrm{P}1 le X le 2 = int_{1}^{2} frac{3}{8}x^2 , mathrm{d}x Find the antiderivative: int frac{3}{8}x^2 , mathrm{d}x = frac{3}{8} cdot frac{x^3}{3} = frac{x^3}{8} Evaluate between the limits x = 1 and x = 2 : left[frac{x^3}{8} right]_{1}^{2} = frac{2^3}{8} - frac{1^3}{8} = frac{8}{8} - frac{1}{8} = frac{7}{8} Common errors include calculating the lower tail mathrm{P}0 le X le 1 = frac{1}{8} , or mistakenly substituting values directly into fx instead of integrating. [M1][A1]

Q8. A curve has equation y = ln2x^3 + 1 . What is the gradient of the tangent to the curve at the point where x = 1 ?

[x] A: 2

[ ] B: 6

[ ] C: dfrac{2}{3}

[ ] D: ln 3

Solution: To find the gradient of the tangent to the curve y = ln2x^3 + 1 at x = 1 , we differentiate y with respect to x using the chain rule. Let u = 2x^3 + 1 , so y = lnu . Then: dfrac{mathrm{d}u}{mathrm{d}x} = 6x^2 quad text{and} quad dfrac{mathrm{d}y}{mathrm{d}u} = dfrac{1}{u} Applying the chain rule: dfrac{mathrm{d}y}{mathrm{d}x} = dfrac{mathrm{d}y}{mathrm{d}u} cdot dfrac{mathrm{d}u}{mathrm{d}x} = dfrac{6x^2}{2x^3 + 1} Now evaluate the derivative at x = 1 : left.dfrac{mathrm{d}y}{mathrm{d}x}right|_{x=1} = dfrac{61^2}{21^3 + 1} = dfrac{6}{3} = 2 Misconception Analysis: - Choice B 6 occurs if one differentiates only the inner function 2x^3 + 1 to get 6x^2 and evaluates it at x = 1 , neglecting the outer logarithmic derivative. - Choice C frac{2}{3} arises from an incorrect derivative of the inner polynomial, such as differentiating 2x^3 incorrectly as 2x . - Choice D ln 3 confuses the gradient of the tangent with the y -coordinate of the point on the curve, y1 = ln21^3 + 1 = ln 3 . [M1][A1]